A RELATIVE PROJ OVER AN INTEGRAL GRADED ALGEBRA IS UNIVERSALLY CLOSED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window 3: statement and proof both lie inside the window 6000-8000 granted to this lane by the owner lane lane-s13 on 2026-10-14.

Editorial note. The statement carries a footnote explaining the integrality hypothesis; the footnote is inside the counted statement span and is preserved verbatim.

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- An integral ring map induces a universally closed morphism. Morphisms, Lemma morphisms-lemma-integral-universally-closed, used at proof lines 7734-7736. Handles Spec(A_0) -> Spec(R).
- Quasi-compactness of Proj of a graded algebra. Constructions, Lemma constructions-lemma-proj-quasi-compact, used at proof lines 7736-7738. Gives quasi-compactness of the first factor.
- Valuative criterion for universal closedness. Constructions, Lemma constructions-lemma-proj-valuative-criterion and Schemes, Proposition schemes-proposition-characterize-universally-closed, used at proof lines 7738-7743. Makes Proj(A) -> Spec(A_0) universally closed.

Source reference. Original lemma 01MD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2118–2118. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01MF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2120–2120. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 035G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4127–4127. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01WM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2487–2487. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original proposition 01KF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2048–2048. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $\mathcal{A}$ be a quasi-coherent graded $\mathcal{O}_S$-algebra. Let $p : X = \underline{Proj}_S(\mathcal{A}) \to S$ be the relative Proj of $\mathcal{A}$. If $\mathcal{O}_S \to \mathcal{A}_0$ is an integral algebra map*[1] *and $\mathcal{A}$ is of finite type as an $\mathcal{A}_0$-algebra, then p is universally closed.*

Proof. The question is local on the base. Thus we may assume that $X = \mathrm{Spec}(R)$ is affine. Let $\mathcal{A}$ be the quasi-coherent $\mathcal{O}_X$-algebra associated to the graded R-algebra A. The assumption is that $R \to A_0$ is integral and A is of finite type over A_0. Write $X \to \mathrm{Spec}(R)$ as the composition $X \to \mathrm{Spec}(A_0) \to \mathrm{Spec}(R)$. Since $R \to A_0$ is an integral ring map, we see that $\mathrm{Spec}(A_0) \to \mathrm{Spec}(R)$ is universally closed, see Morphisms, Lemma 01WM. The quasi-compact (see Constructions, Lemma 01MD) morphism

$$X = \mathrm{Proj}(A) \to \mathrm{Spec}(A_0)$$

satisfies the existence part of the valuative criterion by Constructions, Lemma 01MF and hence it is universally closed by Schemes, Proposition 01KF. Thus $X \to \mathrm{Spec}(R)$ is universally closed as a composition of universally closed morphisms. □

[1] In other words, the integral closure of $\mathcal{O}_S$ in $\mathcal{A}_0$, see Morphisms, Definition 035G, equals $\mathcal{A}_0$.